SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

SMVITM

LearnSphere AI – Personalized Learning Support System

Team Number: 11

Section: AIDS

IDT Prototype Presentation by

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Introduction

- We visited Shri Vishnumoorthi Hayavadanaswami High School to understand students' learning experiences.
- We spoke with students and teachers about difficult subjects, study habits, revision, and classroom challenges.
- The visit helped us identify where simple explanations, visuals, practice questions, and personalized support could improve learning.

Introduction

- Students learn in different ways and at different speeds.
- They prefer simple explanations, real-life examples, visuals, short notes, and practice questions.
- Teachers find it difficult to give every student individual attention within limited classroom time.
- These observations guided us toward a simple, interactive, and personalized learning support system.

Field Visit Details

- **LOCATION: Shri Vishnumoorthi Hayavadanaswami High School , Innanje**
- **DATE OF VISIT:19-09-2026**



Field Visit Details





Stakeholder Details

- **Stakeholders Met**

1.Mrs. Sushma → Headmistress

2. Mrs. Pooja → Teacher (social science)

3. Ananya → student

4.Rahul → student

5.Priya → student

6.Arjun → student

7.Sneha → student

Interview Details

- **Which subjects do you find easy and difficult to understand?**

  Student: I find English is fine, but I find Social science, Mathematics and Science difficult sometimes, especially when the concepts are complicated.

- **What makes a subject difficult for you to learn?**

  Student: Long explanations, difficult words, formulas, and topics that do not have enough examples make it difficult for me to understand.

- **Do you face difficulty in solving Mathematics problems?**

  Student: Yes. Sometimes I understand the formula, but I find it difficult to know which formula or method to use when solving a problem.

- **Do you find it difficult to understand Science concepts without visual explanations?**

  Student: Yes. Diagrams, pictures, videos, and real-life examples make Science

Interview Details

- **Do you find Social Science topics easier when maps, images, or videos are used?**

  Student: Yes. Maps and pictures help me remember places, historical events, and important information better.

- **Do you face difficulties with English grammar or vocabulary?**

  Student: Yes. Sometimes I know the meaning of a sentence but make mistakes in grammar or find it difficult to remember new words.

- **Do you face difficulties in learning Hindi vocabulary or meanings?**

  Student: Yes. I sometimes forget the meanings of Hindi words and need simple explanations or examples to remember them.

Interview Details

- **Do you find it difficult to remember lengthy answers and important concepts?**

  Student: Yes. It is difficult to remember very long answers. Short notes and important points would make revision easier.

- **Do you find it difficult to revise all subjects before examinations?**

  Student: Yes. There are many chapters and subjects, so sometimes I cannot revise everything properly before the examination.

- **How do you usually prepare for your examinations?**

  Student: I mainly study from textbooks, class notes, and previous questions. Before exams, I revise important questions and answers.

- **Do you prefer learning through text, videos, images, or practical examples?**

  Student: I prefer a combination of simple text, images, videos, and practical examples

Interview Details

- **What type of learning content helps you understand a topic faster?**

  Student: Simple explanations, examples, diagrams, short videos, and practice questions help me understand topics faster.

- **Do you get enough opportunity to clear your doubts outside classroom hours?**

  Student: Not always. Sometimes I have doubts after class but may not get enough time to ask the teacher.

- **Would short notes, summaries, practice questions, and quizzes help you during revision?**

  Student: Yes. They would help me revise important topics quickly and check whether I have understood the topic.

Interview Details

- **If you could change one thing about your current learning method, what would it be?**

  Student: I would like more **interactive and visual learning materials** along with simple explanations and practice questions.

- **What are the common learning difficulties you observe among students?**

  Teacher: Students commonly face difficulties in understanding complex concepts, remembering lengthy answers, solving problems, and keeping up with a large syllabus.

- **Which subjects do students generally find more difficult?**

  Teacher: Many students find **Mathematics and Science** difficult. Some students also face challenges with English grammar and Hindi vocabulary.

Interview Details

- **Do students have different learning speeds and learning styles?**

  Teacher: Yes. Every student has a different level of understanding and learning speed. Some students understand quickly, while others need more examples, practice, or visual explanations.

- **What challenges do you face while providing individual attention to every student?**

  Teacher: The main challenge is limited classroom time and the number of students. It is difficult to identify every student's learning gap and provide individual support during regular classes.

- **Would simplified notes, visual explanations, practice materials, and personalized learning support help students?**

  Teacher: Yes. These resources could provide additional support to students, especially for difficult topics, revision, practice, and students who are slow

Key Observation from Field Visit

- Students have different learning abilities, preferences, and learning speeds.
- Students understand concepts better when they are explained using simple language and practical examples.
- Visual content such as images, diagrams, and videos helps students understand difficult concepts more easily.
- Students show greater interest in interactive and activity-based learning compared to only reading textbooks.
- Students face difficulty in remembering lengthy answers and important concepts.
- Students require additional support for revision and examination preparation.
- Different students have different levels of understanding, but classroom teaching may not always provide individual attention to every student.

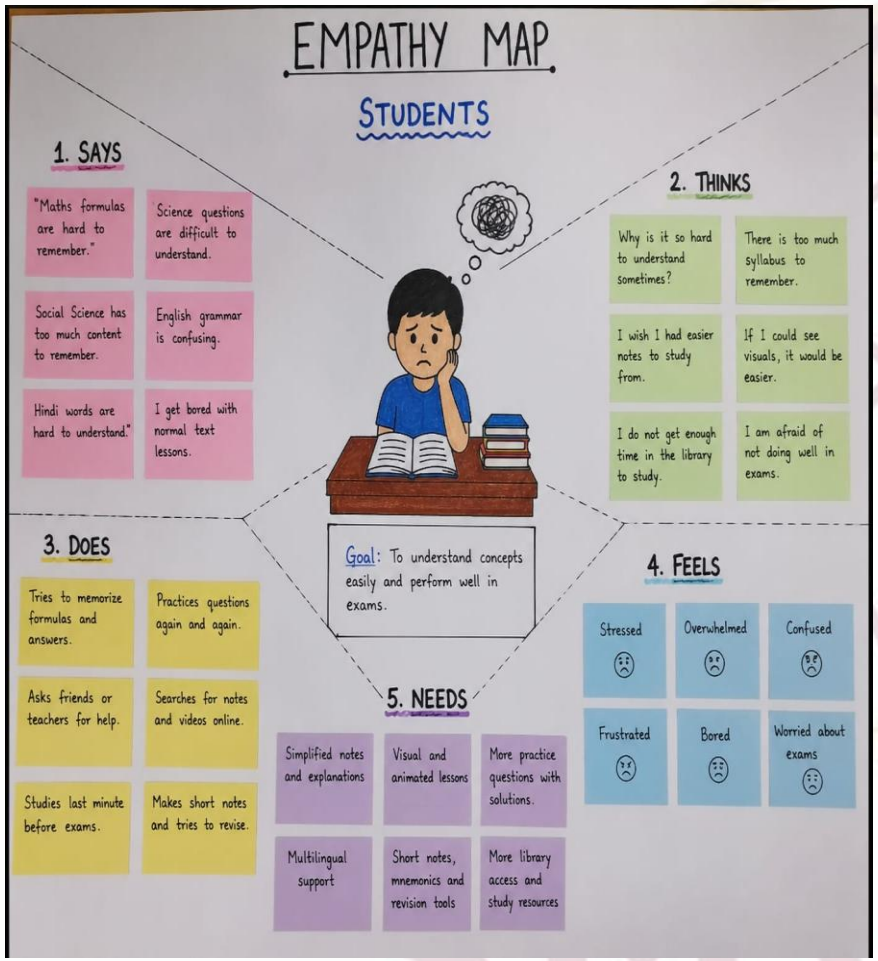
Key Observation from Field Visit

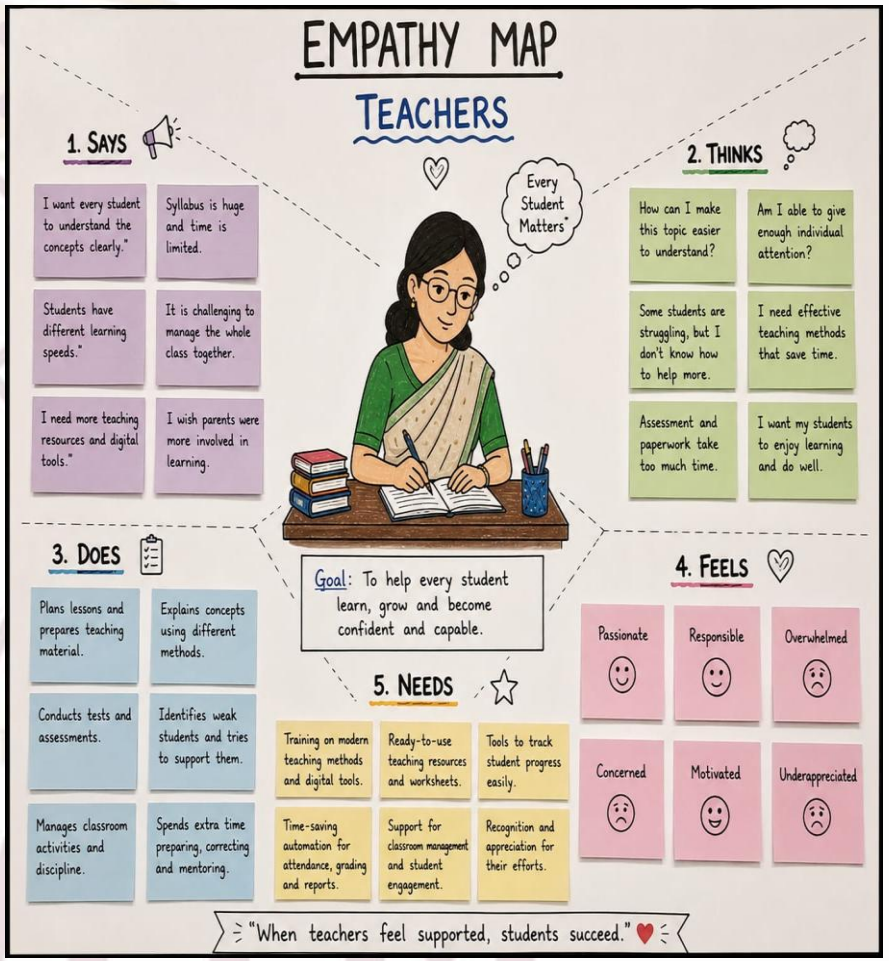
- Students are interested in using digital learning resources when they are simple and easy to access.
- Multilingual learning support can help students understand concepts more comfortably.
- Teachers need effective resources to provide additional learning support and practice materials.
- Students benefit from regular practice, quizzes, examples, and immediate feedback.
- Overall, the field visit highlighted the importance of creating a simple, interactive, accessible, and personalized learning environment.

List of Problems Identified

- Difficulty understanding Mathematics and Science concepts.
- Trouble remembering formulas, vocabulary, and long answers.
- Limited use of visual and interactive learning resources.
- Language barriers in subjects such as English and Hindi.
- Limited personalized support and revision time.

Empathy Map / Problem Tree Analysis





Defined Problem Statement (POV)

- Students often struggle to understand and remember lessons because explanations may be complex, learning resources may be limited, and students learn at different speeds.
- Teachers also have limited time to identify each student's learning gaps and provide individual support.
- LearnSphere AI aims to make learning simpler through personalized explanations, multilingual support, practice questions, and progress tracking.

User Pain Points Identified w.r.t. Defined Problem Statement:

- Understanding complex concepts, especially in Mathematics and Science.
- Choosing the correct formula or method for problems.
- Remembering long answers, formulas, and vocabulary.
- Finding useful diagrams, examples, and visual explanations.
- Revising a large syllabus and clearing doubts outside class.
- Getting enough individual attention and practice feedback.

Techniques/Platforms Used:

1 FRONTEND

HTML, CSS, JavaScript
Student and teacher interface; topic input and dashboards.

2 BACKEND

Python (Flask/FastAPI)
Processes requests and connects the interface, AI tools, and database.

3 AI LEARNING ENGINE

AI / NLP tools
Creates simpler explanations, examples, summaries, and practice questions.

4 PERSONALIZATION

Quiz results + progress rules
Suggests revision and extra practice for topics a student finds difficult.

5 DATA STORAGE

SQLite / MySQL
Stores profiles, quiz scores, learning history, and progress.

6 LANGUAGE SUPPORT

Multilingual NLP / translation
Helps present learning support in English and Kannada, with more languages as feasible.

WORKFLOW Student enters a topic → AI explains it → Quiz & feedback → Progress is tracked → Revision is recommended

Ideas Generated (List 10 or more)

- 1. AI-powered personalized learning
- 2. Learning-gap detection
- 3. Multilingual study assistant
- 4. Interactive digital classroom
- 5. Smart quizzes and instant feedback
- 6. Automatic notes and summaries
- 7. Gamified learning activities
- 8. AI-based career guidance
- 9. Student progress analytics
- 10. Accessible learning resources

Literature Review

Sl.no	Title	Author	Key findings	Year
1	AI-Based Personalized E-Learning Systems: Issues, Challenges, and Solutions	M. Murtaza et al.	Proposes a five-module AI framework. • Covers learner data, adaptive content, recommendations, and assessment. • Highlights personalization challenges and research gaps.	2022
2	Personalized Adaptive Learning Technologies Based on Machine Learning Techniques to Identify Learning Styles: A Systematic Literature Review	S. G. Essa, T. Celik, N. E. Human-Hendricks	• Reviews ML methods for identifying learning preferences. • Finds neural networks are increasingly used. • Notes limited comparisons between deep-learning methods.	2023
3	A Comprehensive Study on Personalized Learning Recommendation in E-Learning System	Q. Bin, M. F. Zuhairi, J. Morcos	• Reviews methods for recommending suitable learning resources. • Highlights learner modelling and recommendation algorithms. • Identifies gaps for future research.	2024
4	Learning Analytics to Support Teaching Skills: A Systematic Literature Review	L. M. Oliva Córdova et al.	• Uses digital learning records to support teaching decisions. • Can inform new teaching methods and online tutoring. • Supports data-informed educational planning.	2021
5	Adaptive Learning Systems: Revolutionizing Higher Education through AI-Driven Curricula	Y. J. J. Alawneh et al.	• Presents an AI-driven adaptive-learning framework. • Uses learner profiles and machine learning to tailor content. • Aims to improve engagement and learning outcomes.	2024

Literature Review contd..

Sl.no	Title	Author	Key findings	Year
6	Personalized Adaptive Learning Platform Empowered by Artificial Intelligence	K. R. Pradeep et al.	• Describes an AI-powered personalized learning platform. • Includes personalized explanations and visual animations. • Discusses webcam-based engagement monitoring.	2024
7	Learning Analytics to Support Education for All: Learning from the Past	M. G. G. Silveira et al.	• Reviews prediction of student outcomes using education data. • Explores educational data mining and learning analytics. • Notes that insights are not always fed back into learning.	2023
8	Trend Intelligent Tutoring System 2018–2022: Systematic Literature Review	J. Suwita et al.	• Reviews intelligent tutoring system components and trends. • Summarizes developments from 2018–2022. • Offers guidance for digital tutoring and self-paced learning.	2024
9	Adaptive Learning Pathways: Advancing Personalization in Online Education Platforms	M. Y. Al Balushi, A. S. Al Harthi	• Reviews personalized learning pathways in online education. • Uses learner interaction and performance data. • Discusses data mining, ML, and NLP techniques.	2024
10	AI-Powered Intelligent Tutoring Systems in Higher Education: A Review of Current Approaches, Challenges, and Future Directions	K. Awham	• Reviews personalized instruction and feedback in tutoring systems. • Discusses current AI-based approaches in higher education. • Highlights transparency, ethics, and scalability challenges.	2025

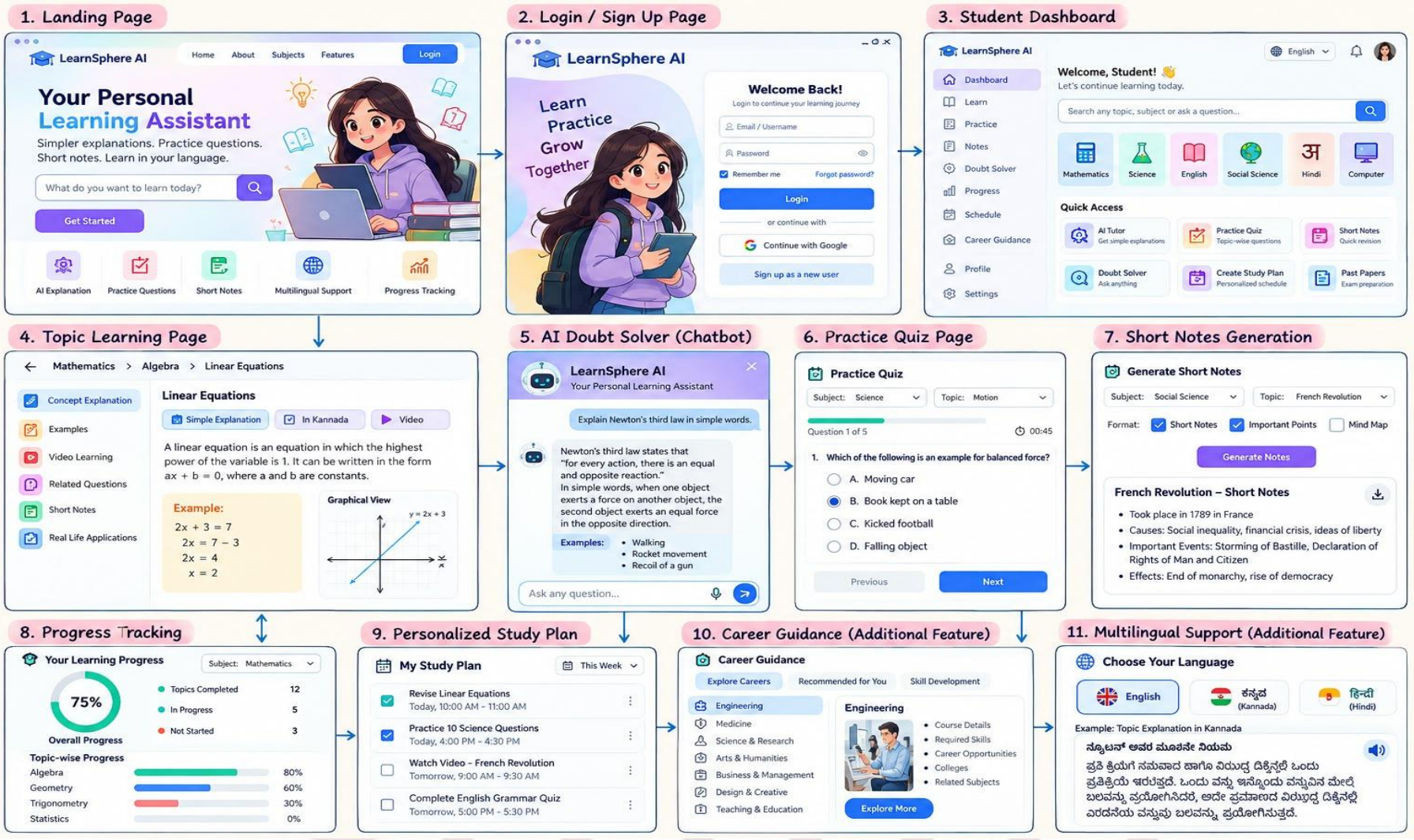
Idea Selected for Prototyping

- Personalized learning: Adjust support to each student's needs and progress.
- Simple explanations: Explain difficult topics using clear language and examples.
- Multilingual support: Help students learn in familiar languages.
- Smart quizzes: Generate practice questions and instant feedback.
- Learning-gap detection: Identify topics that need more practice.
- Study materials: Create short notes, summaries, and worksheets.
- Progress tracking: Show improvement over time.
- Visual learning: Support explanations with diagrams and activities.

Justification for Selection

- Addresses common student learning difficulties.
- Supports both students and teachers.
- Explains difficult topics in simple language.
- Offers multilingual help and practice questions.
- Identifies topics that need more revision.
- Helps teachers prepare learning materials.
- Tracks progress and encourages active learning.

Initial Design Sketch



Sustainable Development Goal (SDG) indicator

SDG 4 – Quality Education

To ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

LearnSphere AI supports inclusive and accessible learning through AI-powered explanations, multilingual assistance, personalized practice, and progress tracking.

Materials Used

1. Frontend: HTML, CSS, JavaScript
2. Backend: Python (Flask/FastAPI)
3. AI Technology: AI and Natural Language Processing (NLP) tools
4. Database: SQLite / MySQL
5. Personalization: Quiz results and progress-tracking rules
6. Language Support: Multilingual NLP and translation tools
7. Learning Resources: Subject notes, diagrams, examples, videos, and practice questions
8. Hardware: Laptop/Desktop and smartphone for accessing and testing the prototype
9. Internet Connection: For online access and AI service integration

Prototype Description

- Student Dashboard: Provides easy access to subjects and learning resources.
- AI Tutor: Explains difficult concepts in simple language with examples.
- AI Doubt Solver: Helps students clarify their doubts.
- Practice Quizzes: Provides questions and instant feedback.
- Short Notes Generator: Creates summaries and important revision points.
- Multilingual Support: Supports learning in English and Kannada.
- Progress Tracking: Monitors student performance and identifies learning gaps.
- Personalized Study Plan: Recommends revision and additional practice.
- Visual Learning: Uses diagrams, examples, and videos to improve understanding.
- Career Guidance: Helps students explore suitable career options.

Photos of Model

- Insert Images

User Testing Report

Feedback Received

- (List out the important/crucial points)

Revisions Made

Final Prototype based on revision

- Add as many slides as required

Conclusion

- Add paragraph/ Points

References

[1] M. Murtaza et al., “AI-Based Personalized E-Learning Systems: Issues, Challenges, and Solutions,” IEEE Access, 2022. doi: 10.1109/ACCESS.2022.3193938.

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- [7] M. G. G. Silveira et al., “Learning Analytics to Support Education for All: Learning from the Past,” IEEE FIE, 2023. doi: 10.1109/FIE58773.2023.10343491.
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- [9] M. Y. Al Balushi and A. S. Al Harthi, “Adaptive Learning Pathways: Advancing Personalization in Online Education Platforms,” IEEE ICCDA, 2024. doi: 10.1109/ICCDA64887.2024.10867273.
- [10] K. Awham, “AI-Powered Intelligent Tutoring Systems in Higher Education: A Review of Current Approaches, Challenges, and Future Directions,” IEEE SCME, 2025. doi: 10.1109/SCME62582.2025.11104905.

Queries?

THANK YOU